Year 8
Mathematics

Unit 2: Equations and Formulae

2.4 Working with Formulae

Working with Formulae

STARTER · 5 MIN

Date: _____ **Title: Working with Formulae**

LEARNING OBJECTIVES

- To be able to work with formulae to solve problems, including those with two unknowns

STARTER *Answer in your book. No calculators.*

1. Solve $3a = 9$
2. Solve $4x = 10$
3. Solve $y + 6 = -4$
4. Solve $2m - 1 = 9$

Starter — Answers

STARTER

Tick or correct in green pen

1. $a = 3$

2. $x = 2.5$

3. $y = -10$

4. $m = 5$

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can substitute values into a formula to calculate an answer	☐
SILVER	I can apply formulae to solve real-life and mathematical problems	☐
GOLD	I can rearrange a simple formula to make a different variable the subject	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Working with Formulae

INTRODUCTION · 3 MIN

A formula links variables. Substitute known values, or rearrange to change the subject.

Substitute

Replace letters with numbers.

$v = u + at$: $u = 3$, $a = 2$, $t = 5 \rightarrow 13$

Solve

Substitute, then solve the equation.

$20 = 4 + 2t \rightarrow t = 8$

Subject

The letter on its own.

v is the subject of $v = u + at$

Rearrange

Use inverse operations.

$P = 4s \rightarrow s = P \div 4$

Examples 1 and 2 — Substituting into Formulae

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint Substitute, then BIDMAS Negatives in brackets

Example 1

$v = u + at$

Find v when $u = 3$, $a = 2$ and $t = 5$.

Example 2

$y = 3x - 6$

Find y when $x = -2$.

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Substitute, then BIDMAS

$v = 3 + 2 \times 5$

$= 3 + 10$

$\mathbf{= 13}$

Tip: *at* means $a \times t$.

Example 2

Use a bracket for the negative

$y = 3 \times (-2) - 6$

$= -6 - 6$

$\mathbf{= -12}$

Tip: $3 \times -2 = -6$.

Task 1

PRACTICE · 9 MIN

BRONZE *Substitute to find the value*

Hint Substitute, then BIDMAS Negatives in brackets

1. $y = 2x - 8,\ x = 1$

2. $y = 5 - 4x,\ x = 1$

3. $y = 3x + 1,\ x = 4$

4. $A = bh,\ b = 6,\ h = 0$

5. $P = 2a + 2b,\ a = 4.5,\ b = 3$

★ $m = \dfrac{a + b}{2},\ a = -3,\ b = 9$

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. $y = -6$

2. $y = 1$

3. $y = 13$

4. $A = 0$

5. $P = 15$

★ $m = 3$

Examples 3 and 4 — Solving with Formulae

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint Substitute, then solve Check your answer

Example 3

$v = u + at$

Find t when $v = 20$, $u = 4$ and $a = 2$.

Example 4

The cost C of n tickets is $C = 12n + 5$.

How many tickets cost \$89?

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

Substitute, then solve

$20 = 4 + 2t$

$16 = 2t$

$\boldsymbol{t = 8}$

Tip: it becomes an equation.

Example 4

Substitute $C = 89$

$89 = 12n + 5$

$84 = 12n$

$\boldsymbol{n = 7}$ **tickets**

Tip: check: 12 × 7 + 5 = 89.

Task 2

PRACTICE · 9 MIN

SILVER *Substitute and solve*

Hint Substitute, then solve Check your answer

1 $P = 4s$

Find s when $P = 30$

2 $v = u + at$

Find a: $v = 25$, $u = 5$, $t = 4$

3 $C = 20 + 15m$

Find m when $C = 95$

4 $A = lw$

Find w when $A = 42$, $l = 6$

5 $F = 1.8C + 32$

Find F when $C = 25$

★ $F = 1.8C + 32$

Find C when $F = 212$

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. $s = 7.5$

2. $a = 5$

3. $m = 5$

4. $w = 7$

5. $F = 77$

★ $C = 100$

Examples 5 and 6 — Changing the Subject

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint Subject: letter on its own Use inverse operations

Example 5

(1 mark)

Make u the subject of

$v = u + at$.

Example 6

(2 marks)

Make x the subject of

$y = 3x + 2$.

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Subtract *at* from both sides

$$v - at = u$$

$$\boldsymbol{u = v - at}$$

Tip: the subject goes on the left.

Example 6

Two inverse steps

$$y - 2 = 3x$$

$$\boldsymbol{x = \frac{y - 2}{3}}$$

Tip: undo + 2, then undo × 3.

Task 3

PRACTICE · 9 MIN

GOLD *Exam-style: rearrange and use*

Hint Subject: letter on its own Use inverse operations

1. $P = 4s$. Make s the subject

2. $y = x + 7$. Make x the subject

3. $A = lw$. Make l the subject

4. $y = 2x - 5$. Make x the subject

5. $x + y = 20$ and $x - y = 4$.
   Find x and y.

★ $2x + 3y = 18$. Give a pair of whole-number solutions.

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1 $s = \frac{P}{4}$

2 $x = y - 7$

3 $l = \frac{A}{w}$

4 $x = \frac{y + 5}{2}$

5 $x = 12, y = 8$

★ e.g. $x = 3, y = 4$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Give a possible solution for $x + y = 20$

SILVER Give a possible solution for $2x + y = 15$

GOLD Give a possible solution for $x - y = 4$

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE **e.g. $x = 12$, $y = 8$**

SILVER **e.g. $x = 5$, $y = 5$**

GOLD **e.g. $x = 7$, $y = 3$**

Exam Practice — Substitution

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint
Substitute, then BIDMAS
Negatives in brackets

Question 1

(2 marks)

$T = 5a + 3b$

Work out T when $a = 4$ and $b = -2$.

$T =$ .....

Question 2

(2 marks)

$v = u + at$

Work out v when $u = 12$, $a = -3$ and $t = 4$.

$v =$ .....

Exam Practice — Using Formulae

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint Substitute, then solve Check your answer

Question 3

(2 marks)

$F = 1.8C + 32$ converts °C to °F.

Work out C when $F = 50$.

..... °C

Question 4

(3 marks)

A hire company uses $C = 35 + 20d$ for d days.

Mia pays $175. For how many days did she hire?

..... days

Exam Practice — Rearranging

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint
Subject: letter on its own
Use inverse operations

Question 5

(2 marks)

Make x the subject of

$y = 5x - 4$

$x =$ .....

Question 6

(2 marks)

$x + y = 15$ and $x - y = 3$

Find the values of x and y.

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

$T = 20 + (-6)$ **M1**

$\mathbf{= 14}$ **A1**

Tip: 3 × −2 = −6.

Question 2

$v = 12 + (-12)$ **M1**

$\mathbf{= 0}$ **A1**

Tip: −3 × 4 = −12.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

$1.8C = 18$ — M1

$\mathbf{C = 10}$ — A1

Tip: subtract 32 first.

Question 4

$175 = 35 + 20d$ — M1

$20d = 140$ — M1

$\mathbf{d = 7}$ — A1

Tip: check: 35 + 140 = 175.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$y + 4 = 5x$ **M1**

$x = \frac{y + 4}{5}$ **A1**

Tip: undo − 4, then ÷ 5.

Question 6

Add: $2x = 18$ **M1**

$x = 9, y = 6$ **A1**

Tip: check both equations.